

Due on tuesday, October 1st, before noon.

Deposit in the MP363 'mailbox' in the physics department.

The questions indicated as [**SELF**] are for self-study and additional exercise.

They will not be marked. (No need to include them in the work that you hand in.) You are however strongly advised to work through them.

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1. (a) [**2+2 pts.**] Look up and report the definitions of
  - the hermitian conjugate of a matrix;
  - a hermitian or self-adjoint matrix.
 (Try any textbook on quantum mechanics or linear algebra, wikipedia,...)
- (b) [**4×2 pts.**] Find out whether the following matrices are hermitian or not. This will involve calculating the hermitian conjugate of each matrix and comparing with the matrix itself.

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad A = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 1 & i \\ 1 & 1 & 0 \\ -i & 0 & 5 \end{pmatrix}$$

- (c) [**SELF**] Look up Pauli matrices. One of the above matrices is a Pauli matrix. Which one, and which Pauli matrix? What do Pauli matrices represent?

2. The equation

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x) \psi(x) = E \psi(x)$$

is the (time-independent) Schroedinger equation for a single particle in a potential  $V(x)$ . Solutions of this equation are stationary states of a particle in the potential  $V(x)$ . Note that  $E$  is a constant (independent of  $x$ ) giving the energy of the stationary state.

- (a) [**2 pts.**] Write down the time-independent Schroedinger equation for the case of no potential,  $V(x) = 0$ .
- (b) [**6 pts.**] Show that the function  $e^{-x^2/2\sigma^2}$  is not a solution of this equation for  $V = 0$ . You can do this by substituting  $\psi(x) = e^{-x^2/2\sigma^2}$  on both sides of the equation, and then showing that the two sides cannot be equal at all  $x$ .

- (c) [5 pts.] Show that the function  $\sin(kx)$  is a solution of this equation with  $V = 0$ , for a specific value of the energy  $E$ . Express this value of the  $E$  in terms of  $k$ ,  $\hbar$  and  $m$ .
- (d) [SELF] Is  $\cos(kx)$  also a solution? Corresponding to the same  $E$  or different  $E$ ?
- (e) [5 pts.] Is the function  $e^{ikx}$  also a solution? Corresponding to the same  $E$  or different  $E$ ?
- (f) [SELF] If  $f_1(x)$  and  $f_2(x)$  are both solutions, corresponding to the same value of  $E$ , then show that any linear combination  $c_1 f_1(x) + c_2 f_2(x)$  is also a solution.  
Does this work if  $f_1(x)$  and  $f_2(x)$  are both solutions, but correspond to different values  $E_1$  and  $E_2$ ?
- (g) [SELF] Can  $e^{ikx}$  be expressed as a linear combination of  $\sin(kx)$  and  $\cos(kx)$ ?
3. (a) [2 pts.] Now write down the time-independent Schroedinger equation for the case of a harmonic potential  $V(x) = \frac{1}{2}m\omega^2 x^2$ .
- (b) [7 pts.] Show that the gaussian function  $e^{-x^2/2\sigma^2}$  is a solution of this equation, for specific values of  $\sigma$  and  $E$ .  
Indicate clearly the constant values of  $\sigma$  and  $E$  for which the gaussian function is a solution. (i.e., express  $\sigma$  and  $E$  as functions of  $\hbar$ ,  $m$  and  $\omega$ , such that  $e^{-x^2/2\sigma^2}$  is a solution)
- (c) [3 pts.] Sketch plots of  $e^{-x^2/2\sigma^2}$  as a function of  $x$ , for  $\sigma = 1$  and  $\sigma = 2$ . Both curves should be on the same graph. Which property of the gaussian curve does  $\sigma$  represent?
- (d) [3 pts.] Does  $\sigma$  increase or decrease with increasing strength of the trapping potential (increasing  $\omega$ )? Explain why your answer makes sense physically.
- (e) [3 pts.] Look up the expression for the energy eigenvalues (energy levels) for the quantum harmonic oscillator. (E.g., on the wikipedia page on “quantum harmonic oscillator”.) Explain how your answer for  $E$  matches one of the energy eigenvalues.
- (f) [2 pts.] Now imagine that the harmonic potential is not centered at zero, but at  $a$ , i.e.,  $V(x) = \frac{1}{2}m\omega^2(x - a)^2$ . Using physical arguments, write down the corresponding gaussian solution to the time-independent Schroedinger equation.